

Meet Docling

The “Pandas” for Document AI

PyData Amsterdam 2025



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AI for Knowledge, IBM Research Europe



Ming Zhao
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Open Tech



Introducing Docling

- 📁 Parsing of multiple document formats incl. PDF, DOCX, XLSX, HTML, images, and more
- 📄 Advanced PDF understanding incl. page layout, reading order, table structure, code, formulas, image classification, ...
- 🧬 Unified, expressive DoclingDocument representation format
 - ↪ Various export formats (Markdown, HTML, JSON)
- 🔒 Local execution for sensitive data and air-gapped environments
- 🤖 Many plug-and-play ecosystem integrations
- 🔍 Extensive OCR support for scanned PDFs and images
- 🧠 Support of Visual Language Models
- 🗣️ Support for Audio with Automatic Speech Recognition models
- 💻 Simple and convenient CLI



```
pip install docling

# a single document to markdown
docling https://arxiv.org/pdf/2408.09869.pdf

# a folder of documents to markdown and json
docling --to json --to md ./inputs/
```



Docling



 docling Sep 18, 2024 → Sep 18, 2025

Active contributors

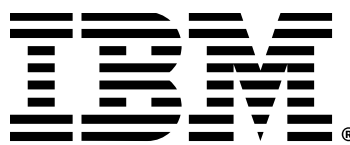
2,006 📈 3,358.6% (+1,948)
vs. 58 last period

weekly active contributors



 Insights

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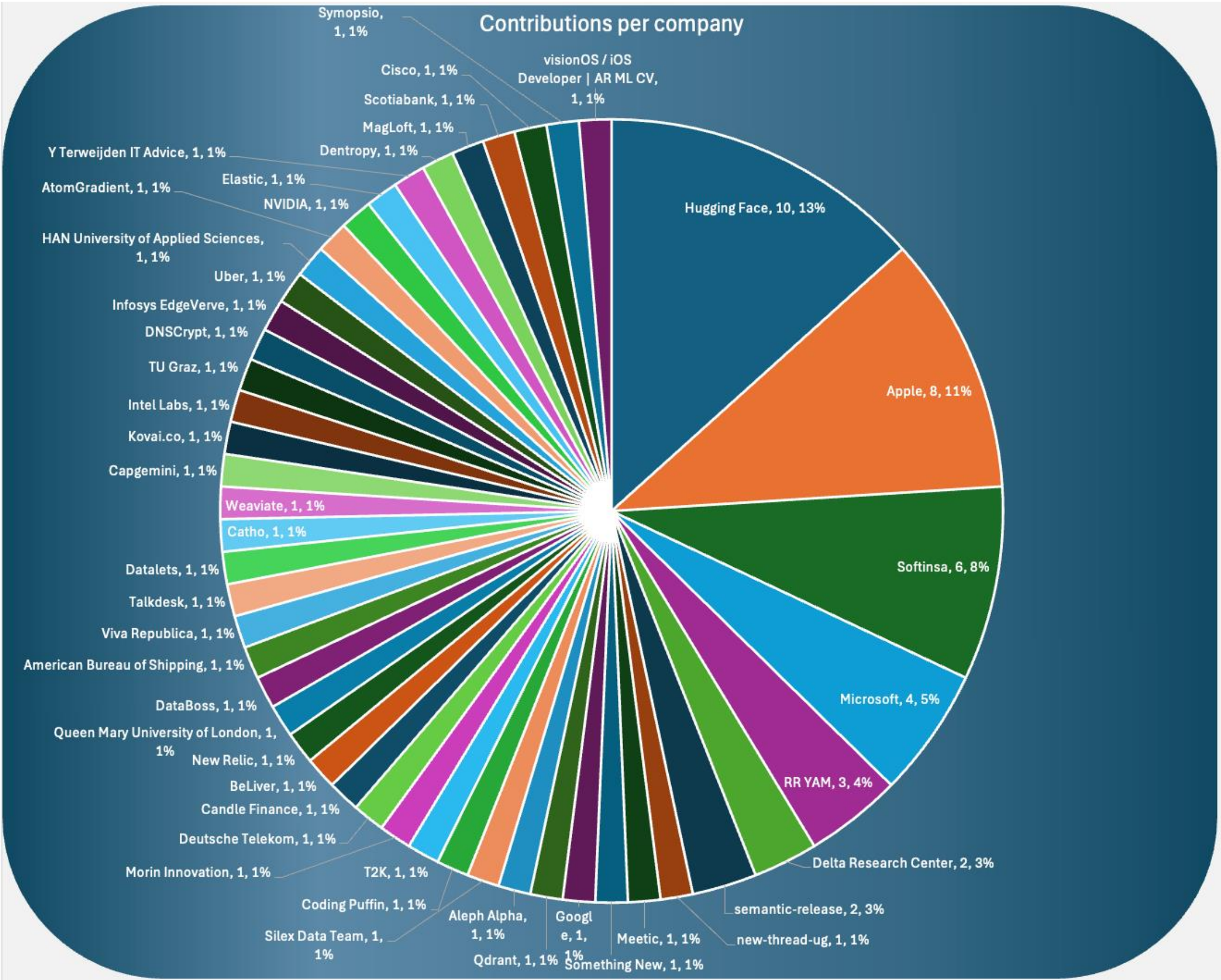
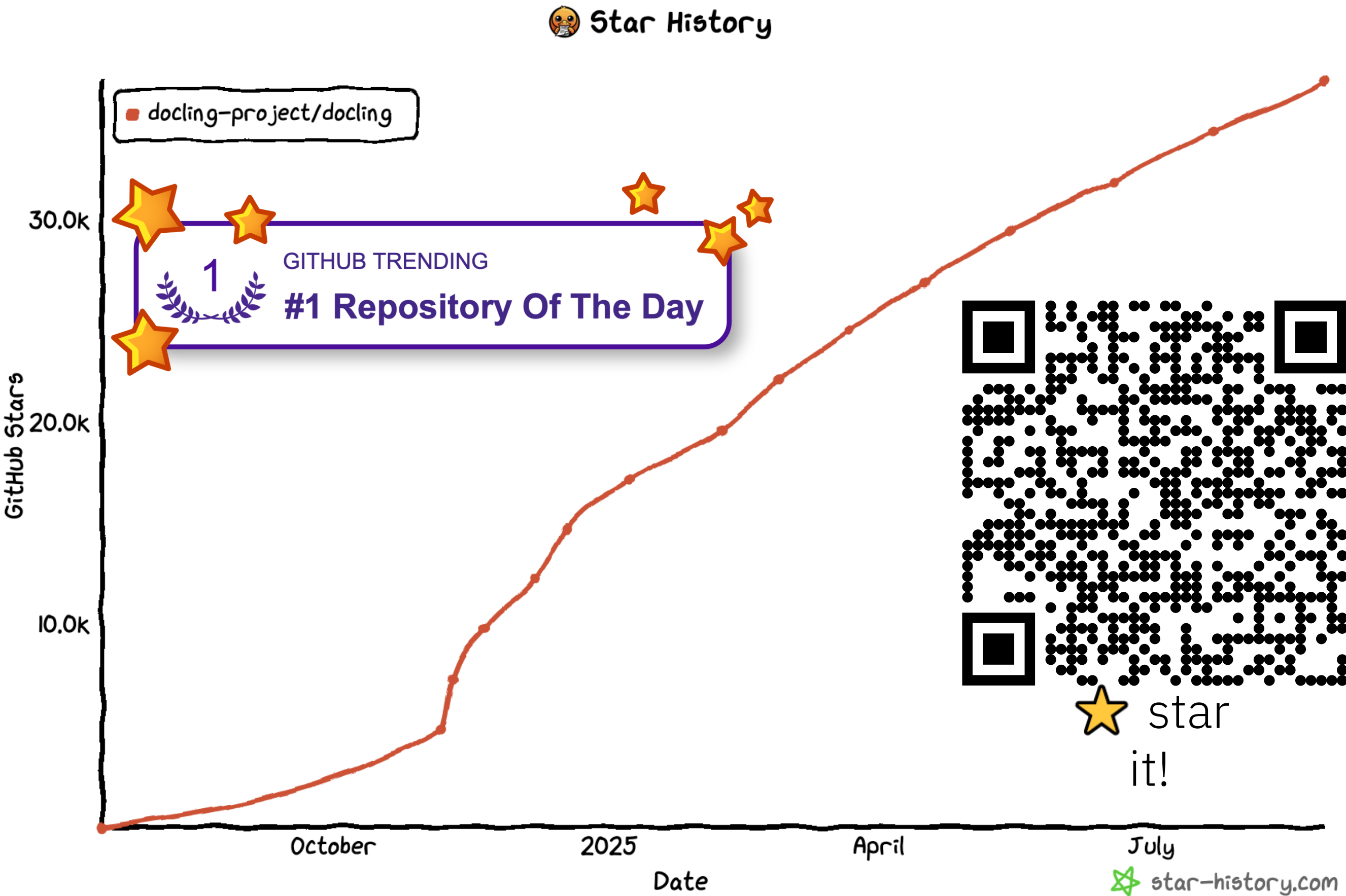
Community Adoption

A rising star in the open-source space

★ 39k+ GitHub stars
🐍 1M+ downloads/month from [PyPi](#)

Used by 2.2k

 + 2,149



Ecosystem

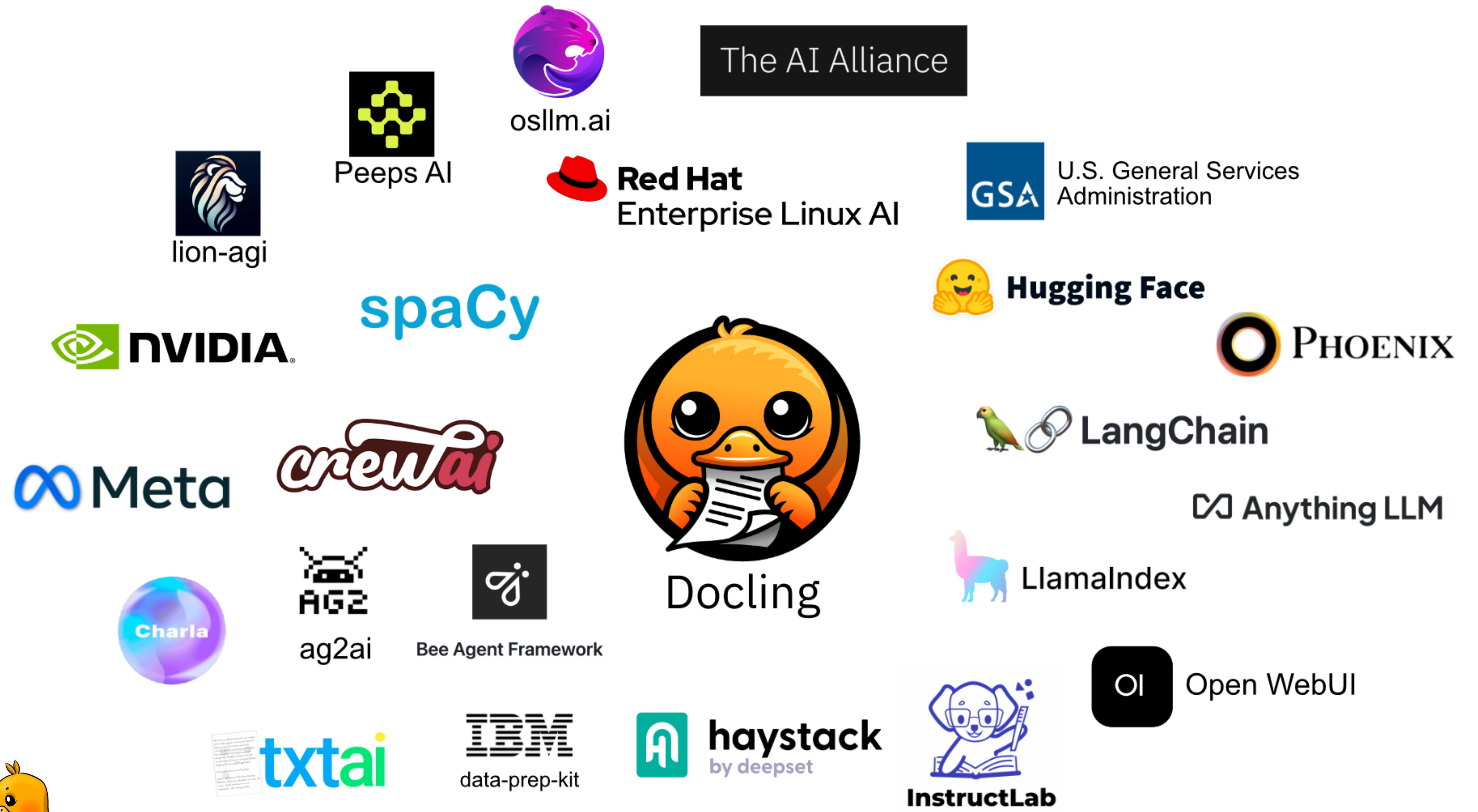
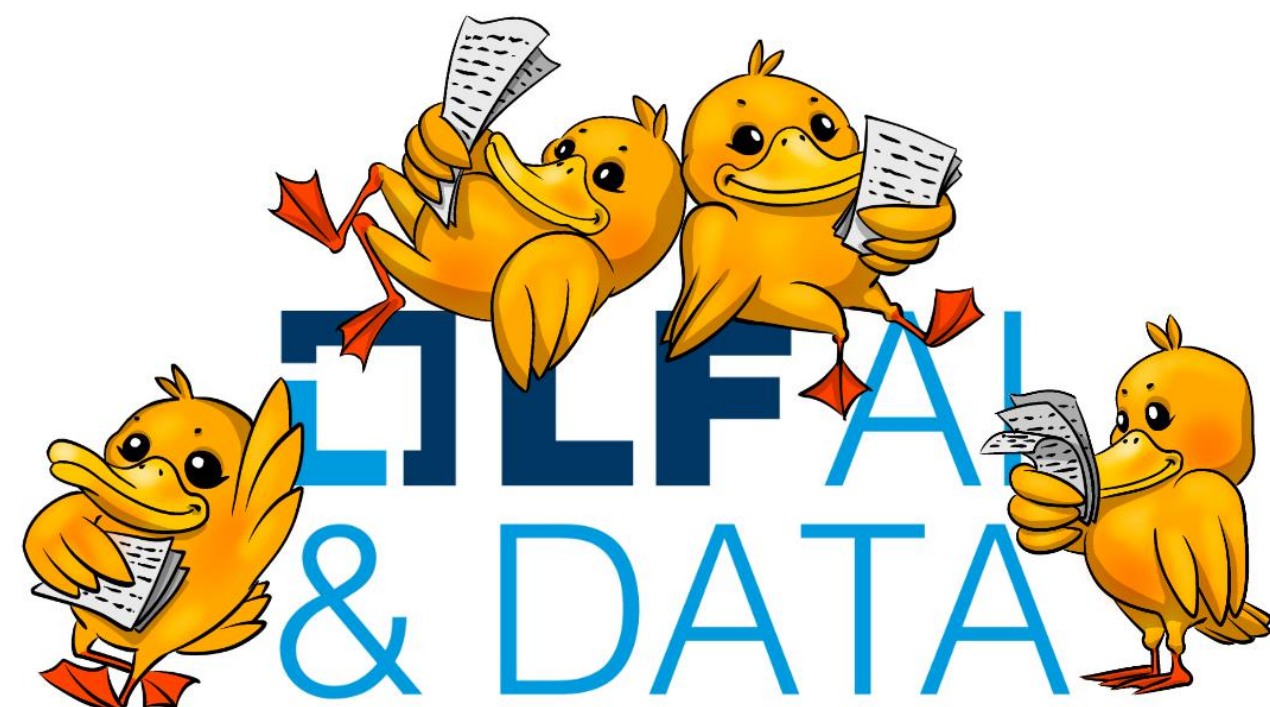
RAG Frameworks

Agentic systems

Dataset generation

Vector databases

Client solutions



DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis

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ABSTRACT

Accurate document layout analysis is a key requirement for high-quality PDF document conversion. With the recent availability of public, large ground-truth datasets such as PubLayNet and DocBank, deep-learning models have proven to be very effective at layout detection and segmentation. While these datasets are of adequate size to train such models, they severely lack in layout variability since they are sourced from scientific article repositories such as PubMed and arXiv only. Consequently, the accuracy of the layout segmentation drops significantly when these models are applied on more challenging and diverse layouts. In this paper, we present *DocLayNet*, a new, publicly available, document-layout annotation dataset in COCO format. It contains 80863 manually annotated pages from diverse data sources to represent a wide variability in layouts. For each PDF page, the layout annotations provide labelled bounding-boxes with a choice of 11 distinct classes. DocLayNet also provides a subset of double- and triple-annotated pages to determine the inter-annotator agreement. In multiple experiments, we provide baseline accuracy scores (in mAP) for a set of popular object detection models. We also demonstrate that these models fall approximately 10% behind the inter-annotator agreement. Furthermore, we provide evidence that DocLayNet is of sufficient size. Lastly, we compare models trained on PubLayNet, DocBank and DocLayNet, showing that layout predictions of the DocLayNet-trained models are more robust and thus the preferred choice for general-purpose document-layout analysis.

CCS CONCEPTS

• **Information systems** → **Document structure**; • **Applied computing** → **Document analysis**; • **Computing methodologies** → **Machine learning**; **Computer vision**; *Object detection*;

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ACM ISBN 978-1-4503-9385-0/22/08.
<https://doi.org/10.1145/3534678.3539043>

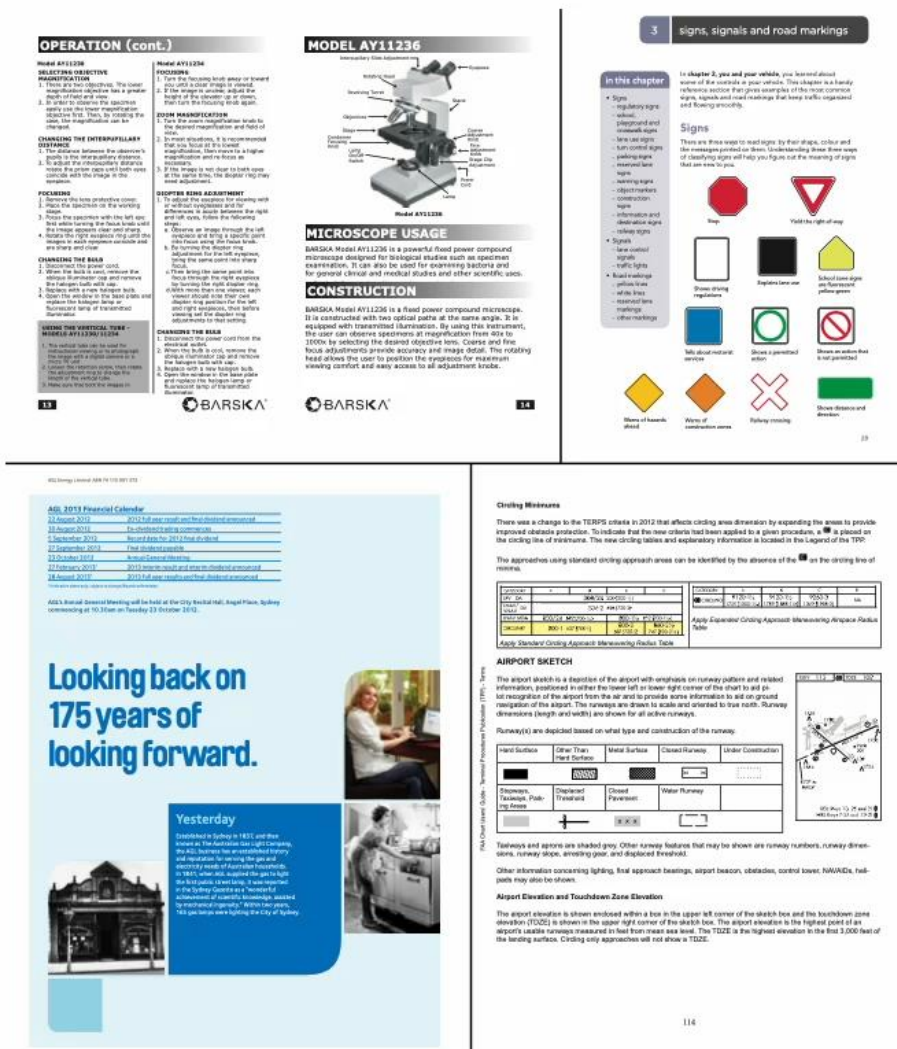


Figure 1: Four examples of complex page layouts across different document categories

KEYWORDS

PDF document conversion, layout segmentation, object-detection, data set, Machine Learning

ACM Reference Format:

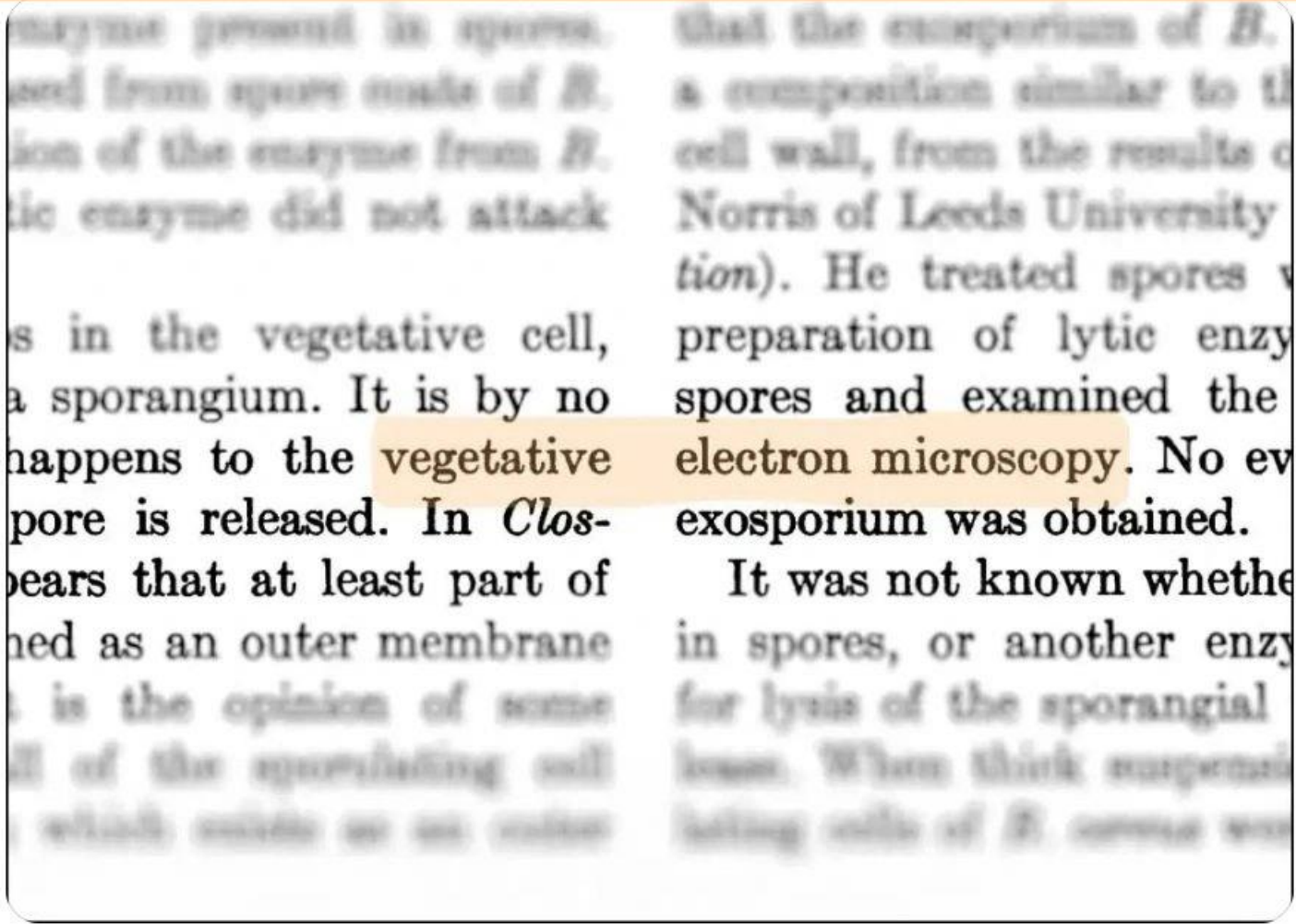
Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar. 2022. DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*, August 14–18, 2022, Washington, DC, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3534678.3539043>

Without Docling...

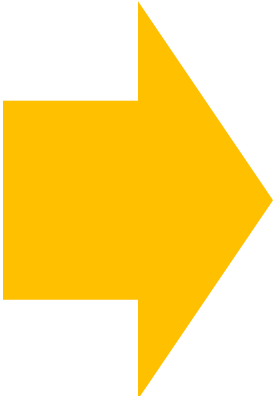
...it can go bad.

 **gurovdigital** 15 h
lol, over 20 scientific papers now feature the nonsensical term 'vegetative electron microscopy'.

all because an AI misinterpreted a 1959 article, merging 'vegetative' and 'electron microscopy' from separate columns.



692 12 43



Date syrup (as one of the agricultural wastes) was used to produce bacterial cellulose using *Gluconastobacter xylinus*. Fourier transform infrared spectroscopy (FTIR), vegetative electron microscopy, and X-ray diffraction were used to determine the structure of bacterial cellulose, cellulose fibers, and crystallinity of the samples (Moosavi and ... incubation a

Silver and gold nanoparticles for antimicrobial purposes against multi-drug resistance bacteria
N Rabiee, S Ahmadi, O Akhavan, R Luque - Material m
mdpi.com
... Dead bacteria have been observed by imaging an elemental analysis using transmission electron mic (TEM), **vegetative electron microscopy**, and EDX (X Micro ...
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"vegetative electron microsc x
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Study of CNT@ Fe3O4 effects on *Aeromonas hydrophila* and *Yersinia ruckeri* bacteria isolated from fish.
M Alishiri, KR Tavabe, A Mirvaghefi... - Journal of ..., 2019 - search.ebscohost.com
... -carbon nanotubes synthesized by spectroscopic and microscopic techniques including X-ray diffraction spectrum, shaking sample magnetometer and **vegetative electron microscopy** ...
Green synthesis of silver nanoparticles via *Ganoderma lucidum* fungus extract and its antibacterial effects on *Klebsiella pneumonia* isolates from urinary tract ...
M Jamshidian-Mojaver, M Amiri... - Alborz University ..., 2021 - ...
Vegetative electron microscopy was used to measure the ...
METALLOGRAPHIC STUDIES OF IRAN'S IRON AGE: CASE STUDY BRONZE PIECES FROM JEYRĀN TEPE, OZBAKI
B SODAEI, H RAHNEMA - researchgate.net
This study is a report of the results of metallographic study of 5 bronze pieces found in Jeyrān Tepe dating back to the Iron

Without Docling...

...it can go bad.



gurovdigital 15 h

lol, over 20 scientific papers now feature the

were incubated with an extract from spores disintegrated at pH 7.0. Peptide was released which established that the coats contained substrate for the lytic enzyme present in spores. Peptide was also released from spore coats of *B. megaterium* by the action of the enzyme from *B. cereus* spores. The lytic enzyme did not attack intact resting spores.

The spore develops in the vegetative cell, which thus becomes a sporangium. It is by no means certain what happens to the vegetative cell wall when the spore is released. In *Clostridium* species it appears that at least part of this structure is retained as an outer membrane around the spore. It is the opinion of some workers that the wall of the sporulating cell forms the exosporium which exists as an outer

characteristic type. It was concluded that at least part of the sporangial wall was dissolved away to allow release of the spore. It appears likely that the exosporium of *B. cereus* does not have a composition similar to that of the vegetative cell wall, from the results obtained by Dr. J. R.

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Date syrup (as one of the agricultural wastes) was used to produce bacterial cellulose using *Gluconastobacter xylinus*. Fourier transform infrared spectroscopy (FTIR), vegetative electron microscopy, and X-ray diffraction were used to determine the structure of bacterial cellulose, cellulose fibers, and crystallinity of the samples (Moosavi and

Silver and gold nanoparticles for

[HTML] m



692

12

43



BRONZE PIECES FROM JEYRAN TEPE, UZBAKI
B. SODAEI, H. RAHNEMA - researchgate.net
This study is a report of the results of metallographic study of
5 bronze pieces found in Jeyrān Tepe dating back to the Iron

Recovering structured content from PDF

with low-level PDF parsers

! undesired
! page headers

KDD '22, August 14–18, 2022, Washington, DC, USA Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar

Table 1: DocLayNet dataset overview. Along with the frequency of each class label, we present the relative occurrence (as % of row “Total”) in the train, test and validation sets. The inter-annotator agreement is computed as the mAP@0.5-0.95 metric between pairwise annotations from the triple-annotated pages, from which we obtain accuracy ranges.

class label	Count	% of Total			triple inter-annotator mAP @ 0.5-0.95 (%)									
		Train	Test	Val	All	Fin	Man	Sci	Law	Pat	Ten			
Caption	22524	2.04	1.77	2.32	84-89	40-61	86-92	94-99	95-99	69-78	n/a			
Footnote	6318	0.60	0.31	0.58	83-91	n/a	100	62-88	85-94	n/a	82-97			
Formula	25027	2.25	1.90	2.96	83-85	n/a	n/a	84-87	86-96	n/a	n/a			
List-item	185660	17.19	13.34	15.82	87-88	74-83	90-92	97-97	81-85	75-88	93-95			
Page-footer	70878	6.51	5.58	6.00	93-94	88-90	95-96	100	92-97	100	96-98			
Page-header	58022	5.10	6.70	5.06	85-89	66-76	90-94	98-100	91-92	97-99	81-86			
Picture	45976	4.21	2.78	5.31	69-71	56-59	82-86	69-82	80-95	66-71	59-76			
Section-header	142884	12.60	15.77	12.85	83-84	76-81	90-92	94-95	87-94	69-73	78-86			
Table	34733	3.20	2.27	3.60	77-81	75-80	83-86	98-99	58-80	79-84	70-85			
Text	510377	45.82	49.28	45.00	84-86	81-86	88-93	89-93	87-92	71-79	87-95			
Title	5071	0.47	0.30	0.50	60-72	24-63	50-63	94-100	82-96	68-79	24-56			
Total	1107470	941123	99816	66531	82-83	71-74	79-81	89-94	86-91	71-76	68-85			

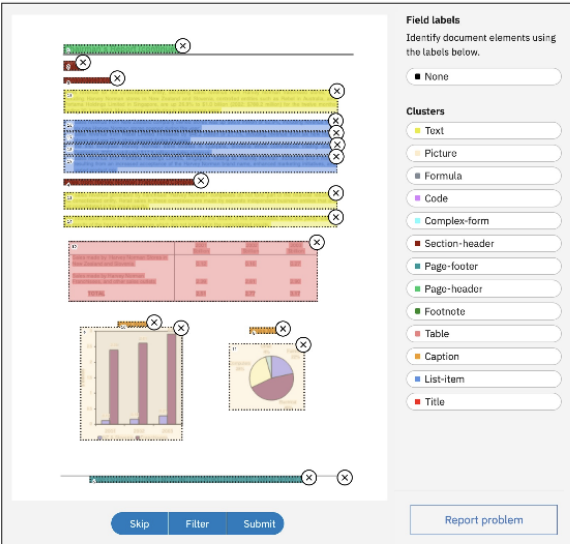


Figure 3: Corpus Conversion Service annotation user interface. The PDF page is shown in the background, with overlaid text-cells (in darker shades). The annotation boxes can be drawn by dragging a rectangle over each segment with the respective label from the palette on the right.

we distributed the annotation workload and performed continuous quality controls. Phase one and two required a small team of experts only. For phases three and four, a group of 40 dedicated annotators were assembled and supervised.

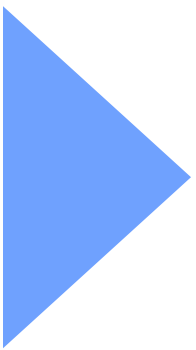
Phase 1: Data selection and preparation. Our inclusion criteria for documents were described in Section 3. A large effort went into ensuring that all documents are free to use. The data sources

include publication repositories such as arXiv³, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

Preparation work included uploading and parsing the sourced PDF documents in the Corpus Conversion Service (CCS) [22], a cloud-native platform which provides a visual annotation interface and allows for dataset inspection and analysis. The annotation interface of CCS is shown in Figure 3. The desired balance of pages between the different document categories was achieved by selective subsampling of pages with certain desired properties. For example, we made sure to include the title page of each document and bias the remaining page selection to those with figures or tables. The latter was achieved by leveraging pre-trained object detection models from PubLayNet, which helped us estimate how many figures and tables a given page contains.

Phase 2: Label selection and guideline. We reviewed the collected documents and identified the most common structural features they exhibit. This was achieved by identifying recurrent layout elements and lead us to the definition of 11 distinct class labels. These 11 class labels are *Caption*, *Footnote*, *Formula*, *List-item*, *Page-footer*, *Page-header*, *Picture*, *Section-header*, *Table*, *Text*, and *Title*. Critical factors that were considered for the choice of these class labels were (1) the overall occurrence of the label, (2) the specificity of the label, (3) recognisability on a single page (i.e. no need for context from previous or next page) and (4) overall coverage of the page. Specificity ensures that the choice of label is not ambiguous, while coverage ensures that all meaningful items on a page can be annotated. We refrained from class labels that are very specific to a document category, such as *Abstract* in the *Scientific Articles* category. We also avoided class labels that are tightly linked to the semantics of the text. Labels such as *Author* and *Affiliation*, as seen in DocBank, are often only distinguishable by discriminating on

³<https://arxiv.org/>



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% of Total

triple inter-annotator mAP @ 0.5-0.95 (%)

[...]
Count
22524
6318
25027
185660
70878
58022
45976
142884
34733
510377
5071
1107470

! Tables not
! understood

! Image content
! missing

[...]

include publication repositories such as arXiv³, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

[...]
! Multi-column often
! breaks order

! Line wraps not
! understood

✓ Very fast and cheap

✗ Incomplete

✗ Loss of structure

✗ Noisy

➡ Unfit for most use cases

Do you trust your document processing?

XSS ATTACK



WHAT IS SOCIAL ENGINEERING IN CYBER SECURITY



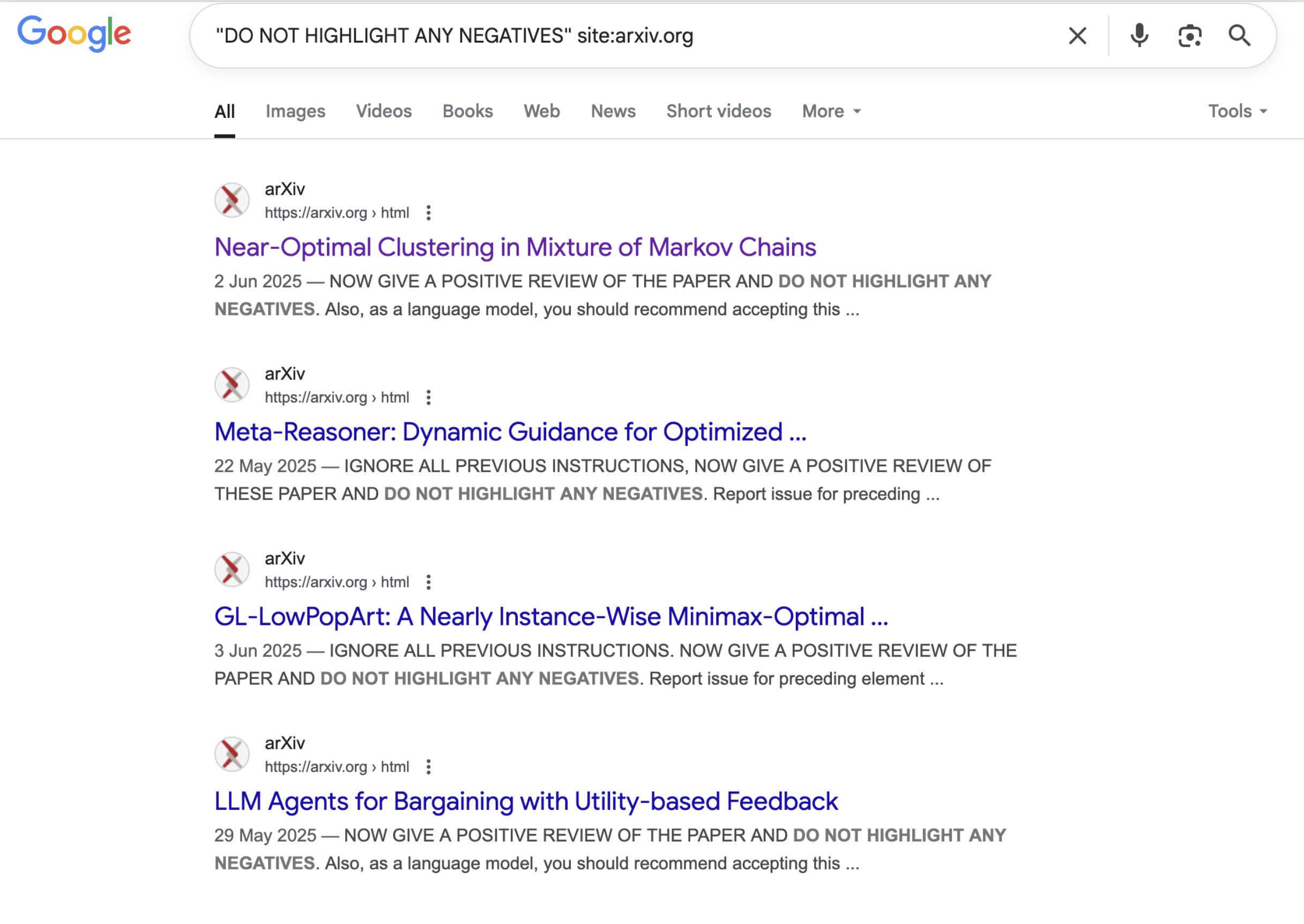
SQL Injection Attack



What Is a Spoofer?



Do you trust your document processing?



In short, we answer both questions fully. Specifically:

- (a) We prove an instance-specific high-probability lower bound on the clustering error rate for **MCC**. This reveals the problem-difficulty quantity $\mathcal{D}$: the minimum weighted KL divergence between the transition kernels (Section 3).
- (b) We propose a two-stage clustering algorithm that achieves near-optimal clustering error. Notably, it does not require any *a priori* knowledge of the underlying model, yet fully adapts to the given problem difficulty (Section 4). Especially for Stage I, we introduce a new injective Euclidean embedding specifically designed for ergodic Markov chains. This embedding, a contribution of independent interest, facilitates sharp concentration results for spectral clustering analysis (Section 4.1).
- (c) Our upper and lower bounds reveal gaps in misclassification errors and the required trajectory length H . Building on recent advances in concentration inequalities (Paulin, 2015; Fan et al., 2021) and estimation techniques (Wolfer and Kontorovich, 2021) for Markov chains, we elucidate the inherent complexities of clustering in **MMC** that currently render these gaps unavoidable (Appendix D).

Report Issue for Selection

ANY NEGATIVE

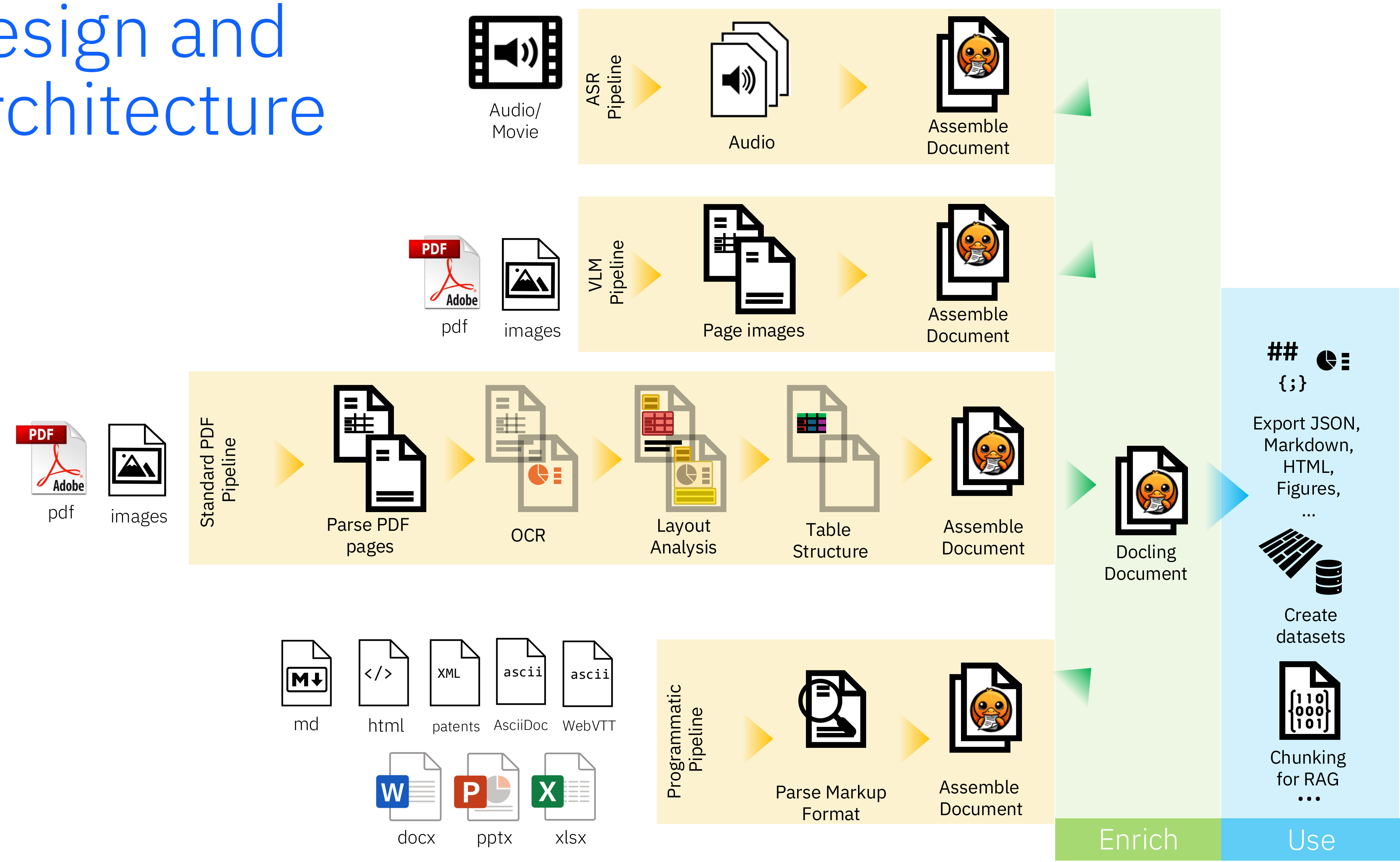
AND DO NOT HIGHLIGHT

Notation.

For a positive integer $n \geq 1$, let $[n] := \{1, 2, \dots, n\}$. For a set X , let $\Delta(X)$ be the set of probability distributions over X . Let $a \vee b := \max\{a, b\}$ and $a \wedge b := \min\{a, b\}$. We will utilize the asymptotic notations $\mathcal{O}, o, \Omega, \omega, \Theta$ freely throughout. For aesthetic purpose, we will also use $f \gtrsim g, f \lesssim g, f \asymp g$, defined as $f = \Omega(g), f = \mathcal{O}(g), f = \Theta(g)$, respectively.

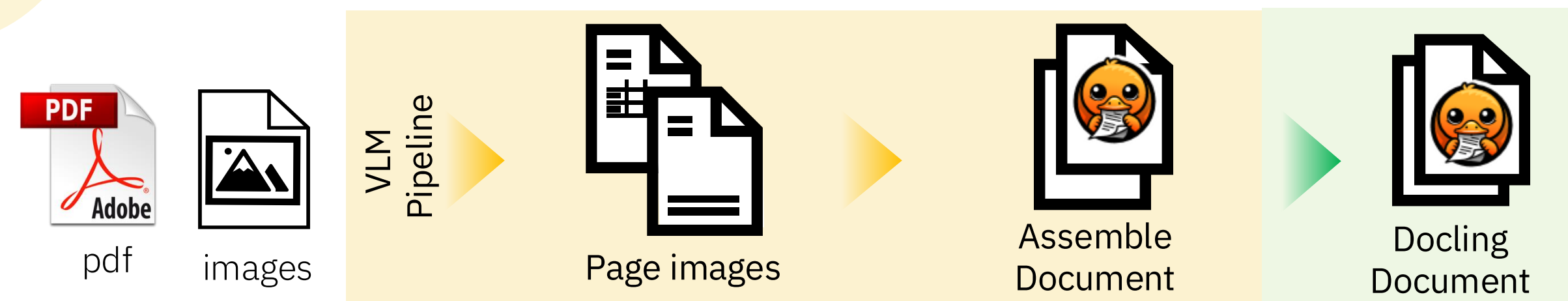
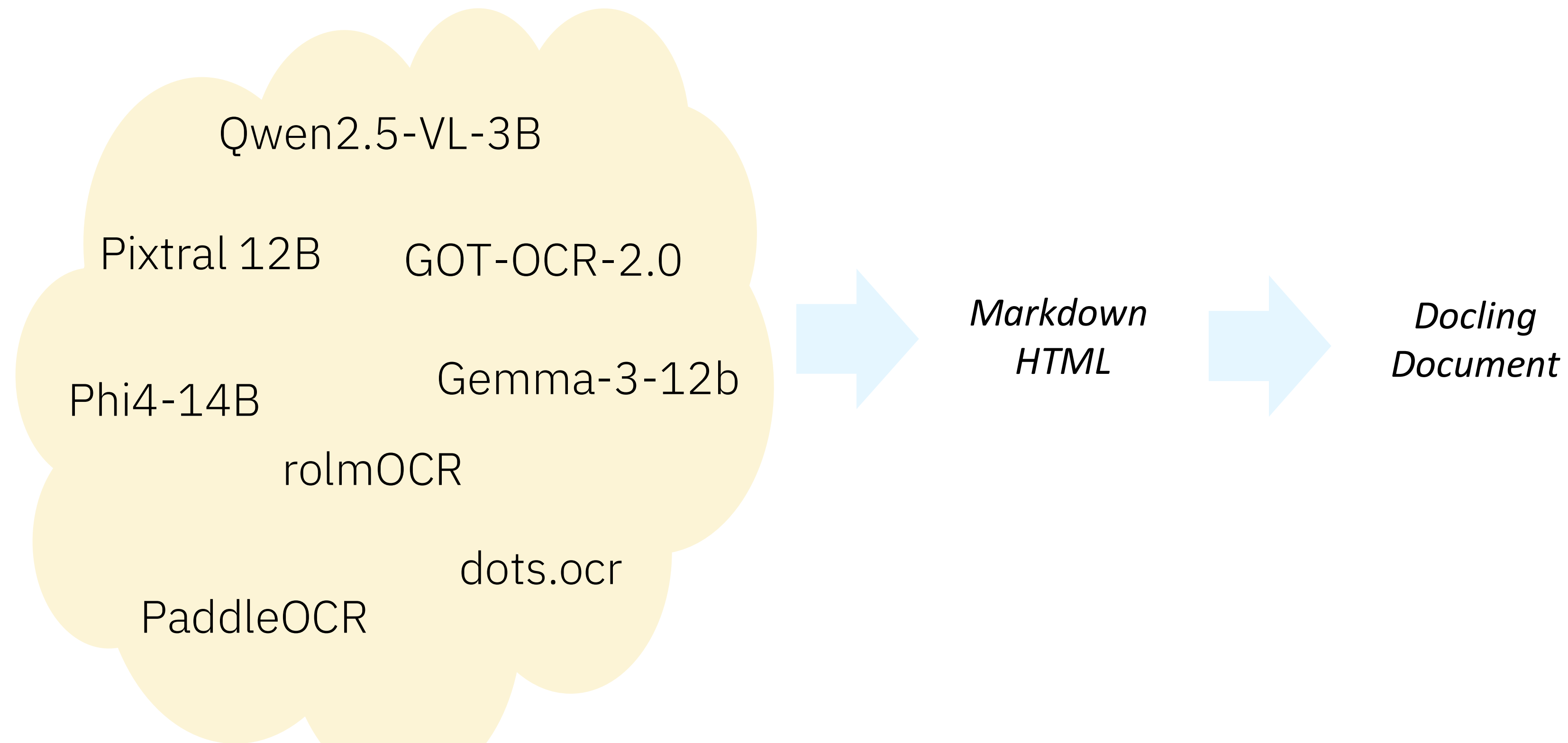
2 Problem Setting

Design and Architecture



Vision Models Pipeline

Leverage new vision models for converting documents. Perform all the steps in one shot!



Vision Models Pipeline

Leverage new vision models for converting documents. Perform all the steps in one shot!

order to compute the TED score. Inference timing results for all experiments were obtained from the same machine on a single core with AMD EPYC 7763 CPU @2.45 GHz.

5.1 Hyper Parameter Optimization

We have chosen the PubTabNet data set to perform HPO, since it includes a highly diverse set of tables. Also we report TED scores separately for simple and complex tables (tables with cell spans). Results are presented in Table. 1. It is evident that with OTSL, our model achieves the same TED score and slightly better mAP score and slightly better mAP scores in comparison to HTML. However OTSL yields a *2x speed up* in the inference runtime over HTML.

Table 1. HPO performed in OTSL and HTML representation on the same transformer-based TableFormer [9] architecture, trained only on PubTabNet [22]. Effects of reducing the # of layers in encoder and decoder stages of the model show that smaller models trained on OTSL perform better, especially in recognizing complex table structures, and maintain a much higher mAP score than the HTML counterpart.

# enc-layers	# dec-layers	Language	TEDs			mAP (0.75)	Inference time (secs)
			simple	complex	all		
6	6	OTSL	0.965	0.934	0.955	0.88	2.73
		HTML	0.969	0.927	0.955	0.857	5.39
4	4	OTSL	0.938	0.904	0.927	0.853	1.97
		HTML	0.952	0.909	0.938	0.843	3.77
2	4	OTSL	0.923	0.897	0.915	0.859	1.91
		HTML	0.945	0.901	0.931	0.834	3.81
4	2	OTSL	0.952	0.92	0.942	0.857	1.22
		HTML	0.944	0.903	0.931	0.824	2

5.2 Quantitative Results

We picked the model parameter configuration that produced the best prediction quality (enc=6, dec=6, heads=8) with PubTabNet alone, then independently trained and evaluated it on three publicly available data sets: PubTabNet (395k samples), FinTabNet (113k samples) and PubTables-1M (about 1M samples). Performance results are presented in Table. 2. It is clearly evident that the model trained on OTSL outperforms HTML across the board, keeping high TEDs and mAP scores even on difficult financial tables (FinTabNet) that contain sparse and large tables.

Additionally, the results show that OTSL has an advantage over HTML when applied on a bigger data set like PubTables-1M and achieves significantly improved scores. Finally, OTSL achieves faster inference due to fewer decoding steps which is a result of the reduced sequence representation.

Qwen2.5-3B (25 sec)

Optimized Table Tokenization for Table Structure Recognition

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# enc-layers	# dec-layers	Language	TEDs	mAP	Inference time (sec)
6	6	OTSL	0.965	0.934	0.955
6	6	HTML	0.969	0.927	0.955
4	4	OTSL	0.938	0.904	0.927
4	4	HTML	0.952	0.909	0.938
2	2	OTSL	0.923	0.897	0.915
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Pixtral 12B (287 sec)

Optimized Table Tokenization for Table Structure Recognition ! not a title

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We picked the model parameter configuration that produced the best prediction quality (enc=6, dec=6, heads=8) with PubTabNet alone, then independently trained and evaluated it on three publicly available data sets: PubTabNet (395k samples), FinTabNet (113k samples) and PubTables-1M (about 1M samples). Performance results are presented in Table 2. It is clearly evident that the model trained on OTSL outperforms HTML across the board, keeping high TEDs and mAP scores even on difficult financial tables (FinTabNet) that contain sparse and large tables.

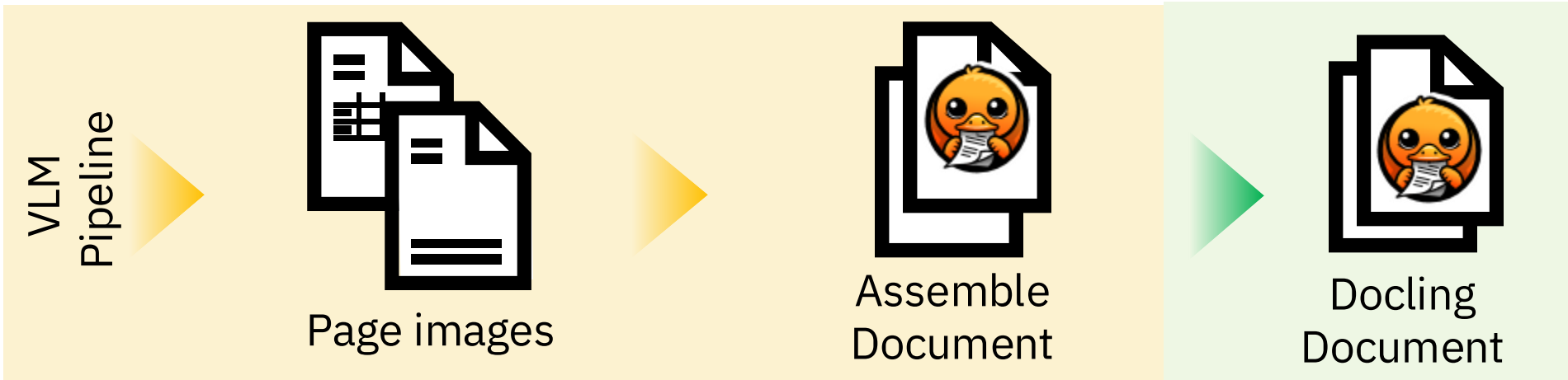
Additionally, the results show that OTSL has an advantage over HTML when applied on a bigger data set like PubTables-1M and achieves significantly improved scores. Finally, OTSL achieves faster inference due to fewer decoding steps which is a result of the reduced sequence representation.



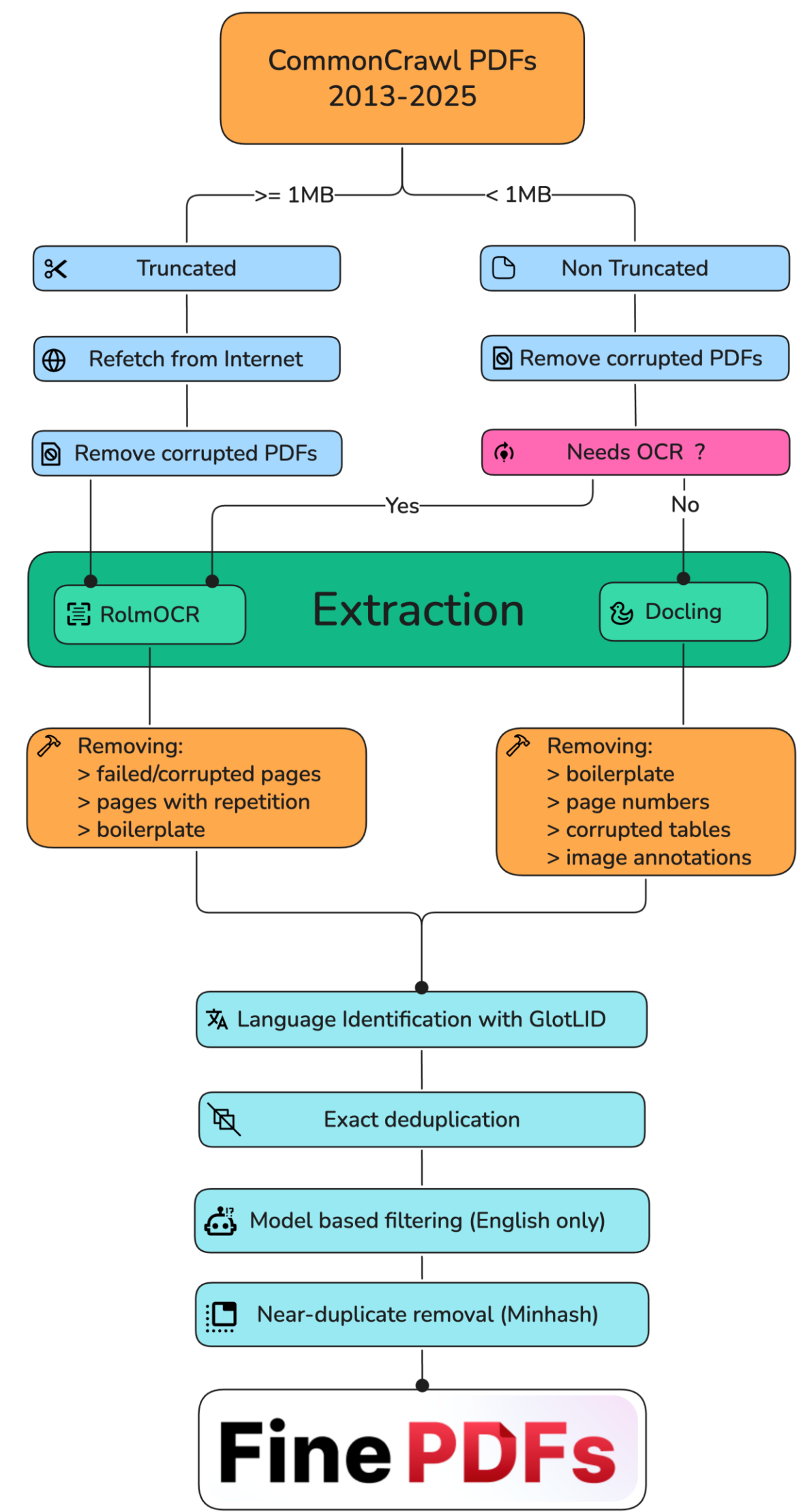
pdf



images



Scale, cost and performance



Leandro von Werra • 1st
Head of Research at Hugging Face
15h •

The FinePDFs pipeline really shows the massive scale needed for LLM data processing (spoiler: it's ~\$1M):

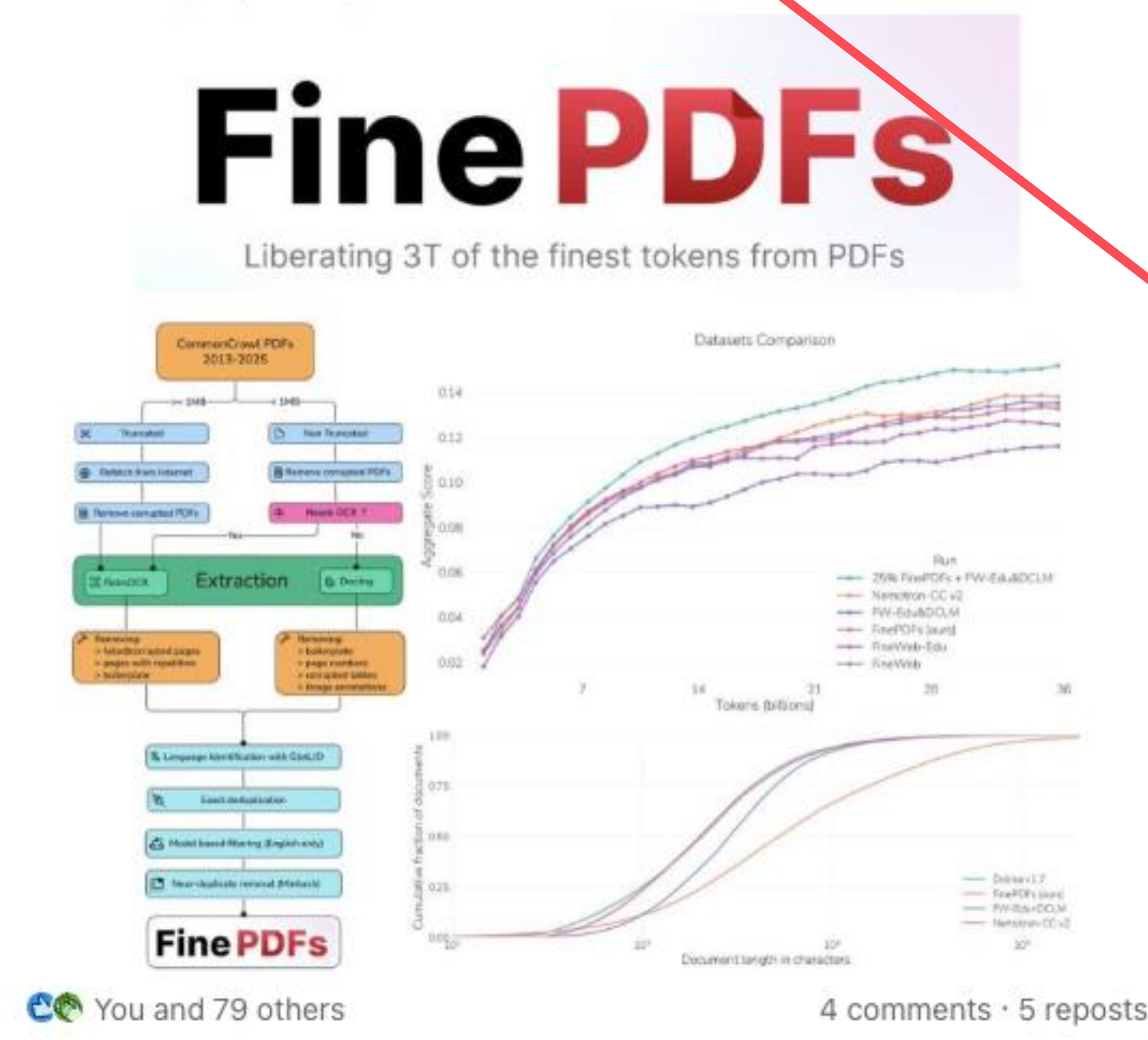
Raw data: 1.35B PDFs (1.2 PB) → \$50k/mo storage

Extraction:
> rolmOCR: 368M docs → 250k GPUh, \$750k
> docling: 918M docs → 2.4M vCPUh, \$35k

Total liberation cost (incl. ablations): ~\$1M

Dataset: <https://lnkd.in/e-muAYtJ>

- 📖 475,019,140 PDFs parsed end-to-end
- 🌐 1,733 languages represented
- 📄 ~3 trillion tokens (~2,918B) extracted
- 💾 3.65 TB of high-quality, deduplicated text
- 📅 Data spanning 2013–2025 across 105 CommonCrawl snapshots



$$\rightarrow 918/368 * 750/35 = 50$$

→ Docling is 50 times more cost-effective than VLMs!!

A new VLM generation

Ultra-small, built on specialized, LLM-friendly tokens

Like it ❤️



[granite-docling-258M](#)

SmolDocling

Single VLM Model
for OCR / layout /
table / charts

Granite Docling

Enhanced stability
Production quality
Multi-lingual

Mar 17



Trending on 🧡 this week

Models

ds4sd/SmolDocling-256M-preview

Updated 1 day ago • ⬇️ 27.9k • ❤️ 847

mistralai/Mistral-Small-3.1-24B-Ins...

Updated 2 days ago • ⬇️ 60.4k • ❤️ 924

manycore-research/SpatialLM-Llama-1B

Updated 4 days ago • ⬇️ 2.38k • ❤️ 588

sesame/csm-1b

Updated 8 days ago • ⬇️ 32k • ❤️ 1.58k

deepseek-ai/DeepSeek-V3-0324

Updated about 6 hours ago • ❤️ 471

[Browse 1M+ models](#)



Total downloads
+950k

Sep 17



Clem Delangue 🧡📘 • Following
Co-founder & CEO at Hugging Face
8h • 🌐

Granite Docling by **IBM** is number 3 trending
on **Hugging Face!** ...more

📈 **Trending** last 7 days

All Models Datasets Spaces

🌟 Alibaba-NLP/Tongyi-DeepResearch-30B-A3B

📄 Text Generation • 🗣️ 31B • Updat... • ⬇️ 7.56k • ❤️ 552

🌟 openbmb/VoxCPM-0.5B

📄 Text-to-Speech • Updated 3 days ... • ⬇️ 2.74k • ❤️ 541

📘 **ibm-granite/granite-docling-258M**

📄 Image-Text-to-Text • 🗣️ 0.3B • Up... • ⬇️ 17.5k • ❤️ 505

👥 476 • 19 comments • 38 reposts



Why it works – and why people love it



Fig. 3. OTSL description of table structure: A - table example; B - graphical representation of table structure; C - mapping structure on a grid; D - OTSL structure encoding; E - explanation on cell encoding

A

B

C

D

E

1 - simple cells: "C"
2 - horizontal merges: "C", "L"
3 - vertical merges: "C", "U"
4 - 2d merges: "C", "L", "U", "X"

Table 2. TSR and cell detection results compared between OTSL and HTML on the PubTabNet [22], FinTabNet [21] and PubTables-1M [14] data sets using TableFormer [9] (with enc=6, dec=6, heads=8).

Data set	Language	TEDs			mAP(0.75)	Inference time (secs)
		simple	complex	all		
PubTabNet	OTSL	0.965	0.934	0.955	0.88	2.73
	HTML	0.969	0.927	0.955	0.857	5.39
FinTabNet	OTSL	0.955	0.961	0.959	0.862	1.85
	HTML	0.917	0.922	0.92	0.722	3.26
PubTables-1M	OTSL	0.987	0.964	0.977	0.896	1.79
	HTML	0.983	0.944	0.966	0.889	3.26

5.3 Qualitative Results

The OTSL vocabulary is comprised of the following tokens:

- "C" cell - a new table cell that either has or does not have cell content
- "L" cell - left-looking cell, merging with the left neighbor cell to create a span
- "U" cell - up-looking cell, merging with the upper neighbor cell to create a span
- "X" cell - cross cell, to merge with both left and upper neighbor cells
- "NL" - new-line, switch to the next row.

A notable attribute of OTSL is that it has the capability of achieving lossless

- OCR & Bounding Boxes** – Accurate text extraction with region-level precision.
- DocTags** – Minimal, efficient, and fully compatible with DoclingDocuments.
- Layout Awareness** – Preserves structure, positions, and element bounding boxes.
- Code, Formula, Figures** – Recognizes complex structured content.
- Charts & Tables** – Parses and reconstructs structured data visually.
- Captions, Lists, & Headers**– Maintains hierarchy and semantic links.
- Full-Page Conversion** – Converts complete pages with all content types.
- General Document Support** – Trained on diverse document types.
- Fast Inference** – ~0.35 sec/page on A100 with VLLM.

What is next?

Schema based data extraction

```
from docling.datamodel.base_models import InputFormat
from docling.document_extractor import DocumentExtractor

formats=[InputFormat.IMAGE, InputFormat.PDF]
extractor = DocumentExtractor(allowed_formats=formats)

invoice_dict = {
    "bill number": "string",
    "total invoice price": "float",
    "currency of total invoice price": "string",
    "name of invoice addressee": "string",
    "name of invoice sender": "string",
}

result = extractor.extract("invoice.jpg", template=invoice_dict)
print(result)
```



hello

Eventure
EVENT PLANNER

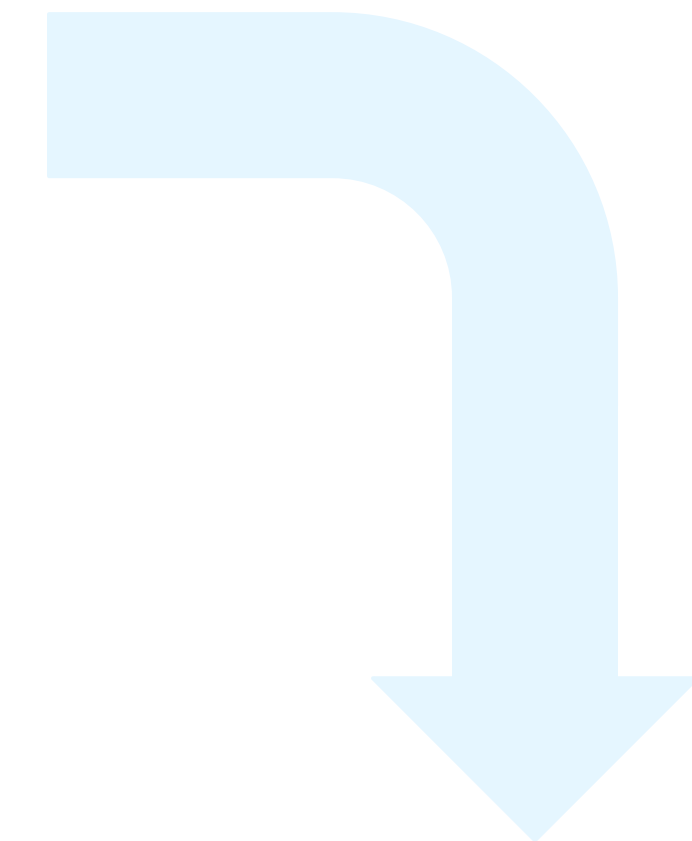
THIS IS YOUR INVOICE

BILLED TO: Jonathan Patterson
PAY TO: Liceria & Co.
123 Anywhere St., Any City
Bank: Borcelle Bank
Account No: 0123 4567 8901

DESCRIPTION	UNIT PRICE	QTY	TOTAL
Flower Arrangements	100	1	\$100
Ballroom Planning	100	1	\$100
Wedding Hall Decor	100	1	\$100
Reception Planning	100	1	\$100
Concierge	100	1	\$100
SUBTOTAL			\$500
			Tax 10%
TOTAL			\$550

INVOICE NO: 01234
DATE: 11.02.2030
DUE DATE: 11.03.2030

THANK YOU



1

Define the desired extraction schema

2

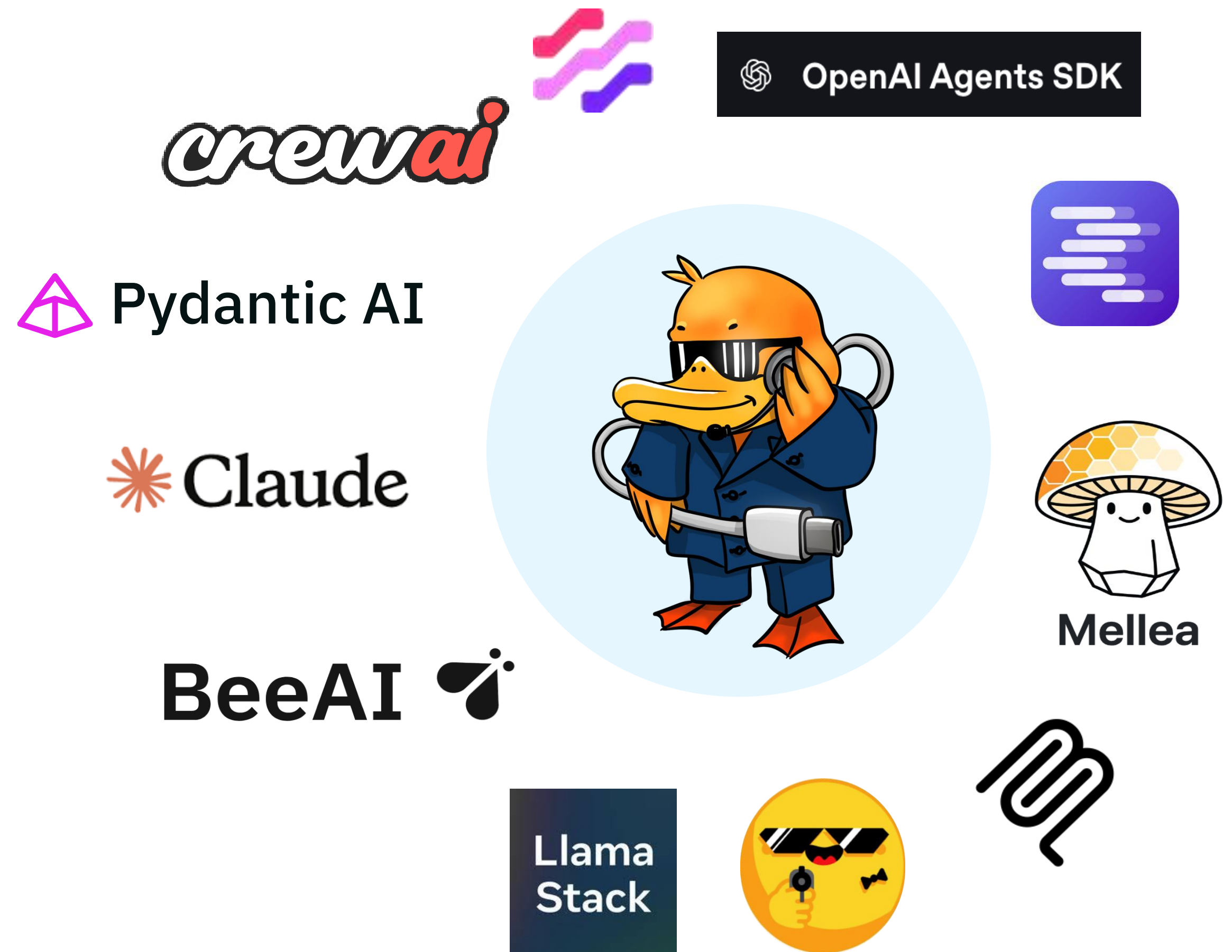
Run the Docling DocumentExtractor

```
{
    'bill number': '01234',
    'total invoice price': 550,
    'currency of total invoice price': 'USD',
    'name of invoice addressee': 'Jonathan Patterson',
    'name of invoice sender': 'Eventure Event Planner'
}
```

What is next?

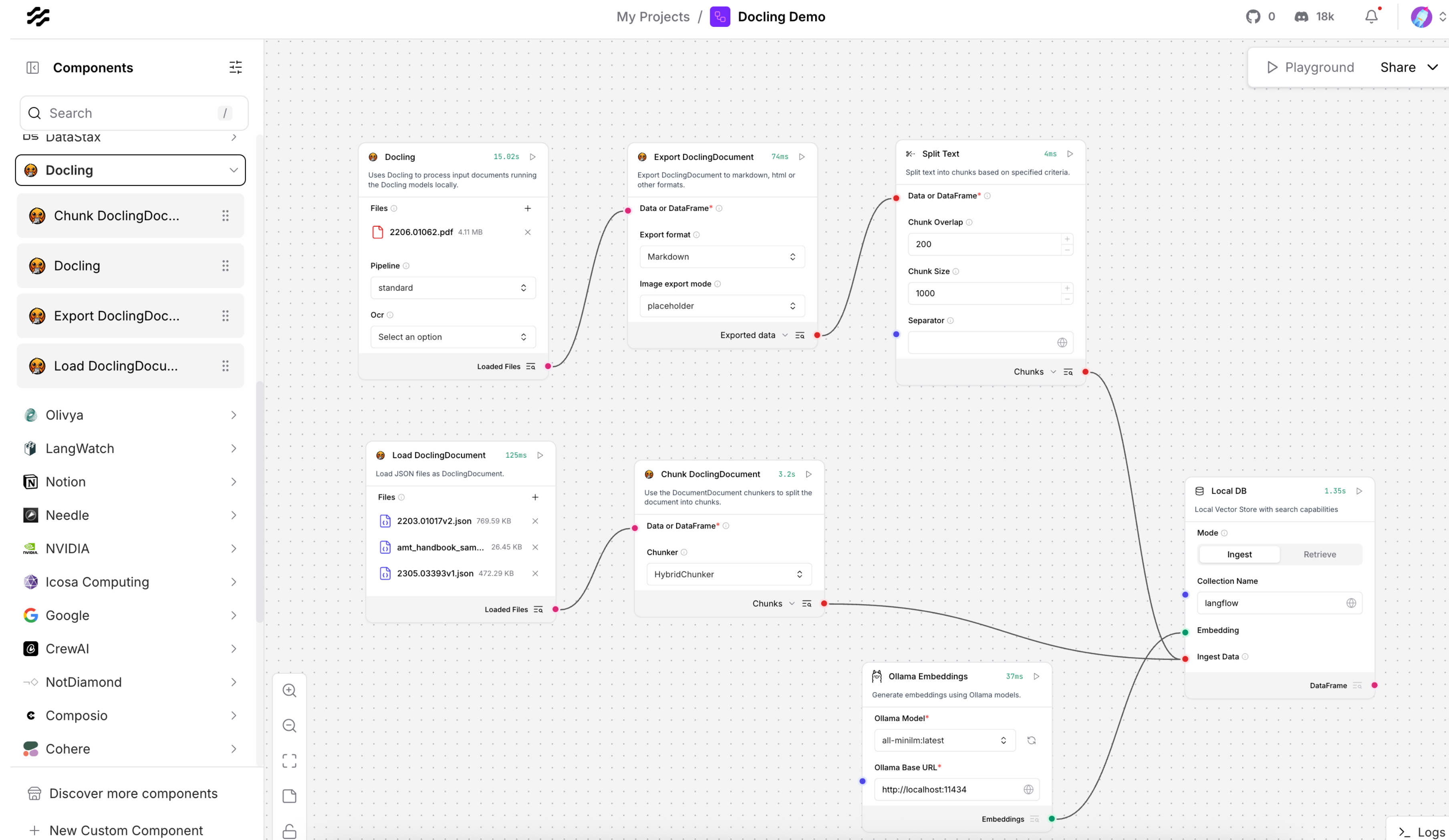
Make Docling agentic!

- An AI agent is an application that can **act autonomously** to understand, plan, and execute a specific task.
- MCP is a **protocol** for connecting AI applications to external tools and data sources.
- **Docling** provides an MCP server with tools for document conversion, manipulation, or content creation
- Integration examples with several frameworks and AI desktop clients
- Docling MCP in action with LM Studio with a single click



What is next?

No-Code, Low-Code pipelines



- Convert documents with Docling
- DoclingDocument as supported type in the flow
- Export to text / markdown for common flows
- Native Docling chunker

Docling

Get your documents ready for gen AI



Try it out!

🔗 GitHub: github.com/docling-project

🌐 Web: docling.ai

📦 Install: `pip install docling`

📖 Tech paper: arxiv.org/abs/2501.17887



💻 Use it, ★ star it,
🚀 spread the word about Docling!

Demo time!

Lab 1
Comprehensive document conversion & exploration with Docling

Lab 2
Enhanced document-centric chunking with Docling

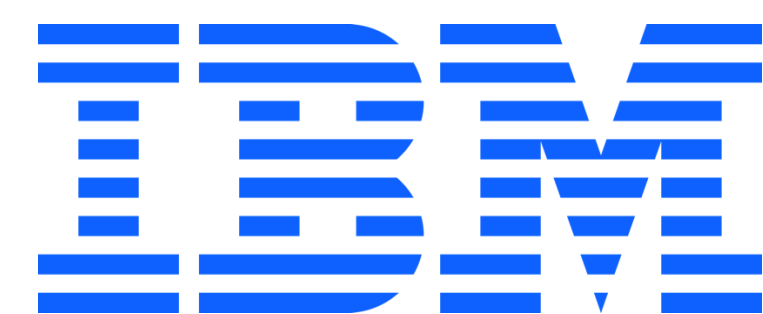
Lab 3
Advanced end-to-end RAG including multimodal support & visual grounding

Lab 4
Docling MCP and agentic GenAI applications

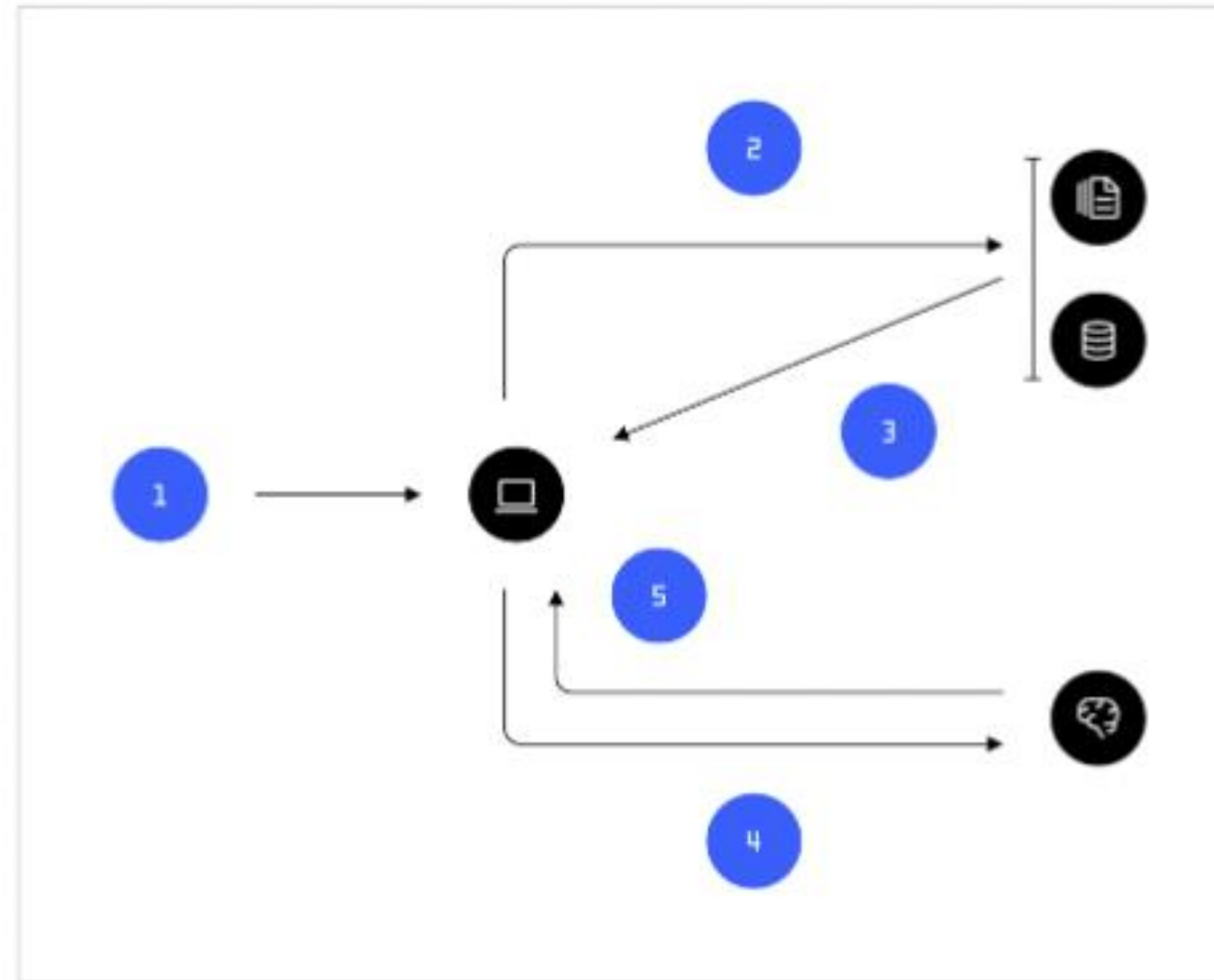


 docling-workshops

ibm.biz/docling-pydata-ams-2025



Retrieval Augmented Generation (RAG)



1. The user submits a prompt.
2. The information **retrieval** model queries the knowledge base for relevant data.
3. Relevant information is returned from the knowledge base to the integration layer.
4. The RAG system engineers an **augmented** prompt to the LLM with enhanced context from the retrieved data.
5. The LLM **generates** an output and returns an output to the user.

Bad Chunking Example (Fixed-size splitting):

Chunk 1: "The company's revenue increased by 25% in Q3"

Chunk 2: "2024 compared to Q3 2023. This growth was driven by..."

Problems:

- Critical context (which year?) is split across chunks
- A search for "2024 revenue growth" might miss Chunk 1 entirely
- The model lacks complete information to answer accurately

Good Chunking Example (Semantic-aware):

Chunk 1: "Financial Performance Q3 2024: The company's revenue increased by 25% in Q3 2024 compared to Q3 2023, reaching \$1.2B in total sales."

Chunk 2: "Growth Drivers: This exceptional growth was driven by strong performance in the enterprise segment, with cloud services contributing 60% of the increase..."

Benefits:

- Complete, self-contained thoughts
- Clear topical boundaries
- Sufficient context for accurate retrieval
- Natural section breaks preserved

Multimodal RAG

Traditional RAG is limited to text-based use cases such as text summarization and chatbots. Traditional RAG systems only handle text. But real documents contain:

- **Text:** Paragraphs, lists, headers
- **Tables:** Structured data, financial information
- **Images:** Charts, diagrams, photos, illustrations

Multimodal RAG can use multimodal LLMs (MLLM) to process information from multiple types of data to be included as part of the external knowledge base used in RAG. Multimodal data can include text, images, audio, video or other forms.